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Inventor(s)	Ghannam; Mahmoud Yousef et al.

Detection system for a vehicle

Abstract

A vehicle includes a plurality of RADAR modules arranged at a common side of a plurality of sides of the vehicle and configured to detect an obstruction in a region exterior to the vehicle. A first RADAR module is on a left portion of the common side and has a first field of view. A second RADAR module is on a right portion of the common side and has a second field of view. A third RADAR module is between the first and second RADAR modules and is spaced from the first RADAR module by a first distance and spaced from the second RADAR module by a second distance. The third RADAR module has a third field of view at least partially overlapping each of the first and second fields of view. Control circuitry triangulates a position of the obstruction between a pair of the plurality of RADAR modules.

Inventors: Ghannam; Mahmoud Yousef (Canton, MI), Van Wiemeersch; John Robert (Novi, MI), Elangovan; Vivekanandh (Canton, MI), Colella; Nicholas (Grosse Ile, MI)

Applicant: Ford Global Technologies, LLC (Dearborn, MI)

Family ID: 93382194

Assignee: Ford Global Technologies, LLC (Dearborn, MI)

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Primary Examiner: Huynh; Hai H

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Background/Summary

FIELD OF THE DISCLOSURE

(1) The present disclosure generally relates to a detection system for a vehicle and, more particularly, to detection techniques for a vehicle using an arrangement of time-of-flight sensors.

BACKGROUND OF THE DISCLOSURE

(2) Conventional exterior detection systems for vehicles may incorporate time-of-flight technology.

SUMMARY OF THE DISCLOSURE

(3) According to a first aspect of the present disclosure, a vehicle includes a plurality of sides each having a left portion and a right portion, the plurality of sides including a front side, a rear side, a left side, and a right side. A plurality of RADAR modules arranged at a common side of the plurality of sides and configured to detect an obstruction in a region exterior to the vehicle. The plurality of RADAR modules includes a first RADAR module on the left portion of the common side and having a first field of view. The plurality of RADAR modules further includes a second RADAR module on the right portion of the common side and has a second field of view. The plurality of RADAR modules further includes a third RADAR module between the first and second RADAR modules and spaced from the first RADAR module by a first distance and spaced from the second RADAR module by a second distance, the third RADAR module has a third field of view at least partially overlapping with each of the first and second fields of view. Control circuitry is in communication with the plurality of RADAR modules and configured to triangulate a position of the obstruction between a first pair of the plurality of RADAR modules.

(4) Embodiments of the first aspect of the present disclosure can include any one or a combination of the following features: the first distance is between a quarter of and four times the second distance; the first distance is between a half of and twice the second distance; the third RADAR module is vertically spaced from the first and second RADAR modules, and the first RADAR module is vertically aligned with the second RADAR module; a first taillight assembly disposed at the left portion of the rear side, and a second taillight assembly disposed at the right portion of the rear side, wherein the first and second RADAR modules are disposed in the first and second taillight assemblies, respectively; the control circuitry is configured to selectively execute a blind-spot detection algorithm to determine at least one object in a blind spot for the vehicle based on the positional data captured in the first and second fields of view, and a cross-traffic detection algorithm to determine the at least one object in a cross-traffic area based on the positional data captured in the first and second fields of view; a powertrain of the vehicle in communication with the control circuitry, wherein the selective execution of the blind-spot detection algorithm and the cross-traffic detection algorithm is based on a gear status of the powertrain; the control circuitry is configured to execute at least one of the blind-spot detection algorithm and the cross-traffic detection algorithm when the gear status is not park; the control circuitry is further configured to execute a parking assist algorithm that guides steering of the vehicle based on positional data captured in both the third field of view and at least one of the first and second fields of view; the control circuitry is configured to select between the parking assist algorithm and the blind-spot detection algorithm based on the gear status; a tailgate at the rear side of the vehicle, wherein the third RADAR module is operably coupled with the tailgate; the control circuitry is configured to execute a gesture recognition algorithm that determines a gesture of a user outside of the vehicle based on the positional information captured in the third field of view; the control circuitry is configured to execute the gesture recognition algorithm when the gear status is park; and the control circuitry is configured to triangulate the position of the obstruction between a second pair

of the plurality of the RADAR modules, compare the triangulation based on the first pair of the plurality of RADAR modules to the triangulation based on the second pair of the plurality of RADAR modules, and update the position based on the comparison.

(5) According to a second aspect of the present disclosure, a vehicle includes a front, a rear, a driver side, and a passenger side. The vehicle further includes a plurality of RADAR modules arranged at the rear of the vehicle and configured to detect an obstruction in a region exterior to the vehicle. The plurality of RADAR modules includes a first RADAR on the driver side of the vehicle and having a first field of view. The plurality of RADAR modules further includes a second RADAR module on the passenger side of the vehicle and having a second field of view. The plurality of RADAR modules further includes a third RADAR module between the first and second RADAR modules and spaced from the first RADAR module by a first distance and spaced from the second RADAR module by a second distance, the third RADAR module having a third field of view at least partially overlapping with each of the first and second fields of view behind the vehicle. The plurality of RADAR modules further includes control circuitry in communication with the plurality of RADAR modules and configured to triangulate a position of the obstruction between a first pair of the plurality of RADAR modules, wherein the first distance is between a half of and twice the second distance.

(6) Embodiments of the second aspect of the present disclosure can include any one or a combination of the following features: the control circuitry is configured to triangulate the position of the obstruction between a second pair of the plurality of the RADAR modules; a first taillight assembly disposed at the rear on the driver side, and a second taillight assembly disposed at the rear on the passenger side, wherein the first and second RADAR modules are disposed in the first and second taillight assemblies, respectively; the control circuitry is configured to selectively execute a blind-spot detection algorithm to determine at least one object in a blind spot of at least one of the driver side and the passenger side based on positional data captured in the first and second fields of view, and a cross-traffic detection algorithm to determine the at least one object in a cross-traffic area based on the positional data captured in the first and second fields of view; and a powertrain of the vehicle in communication with the control circuitry, wherein the selective execution of the blind-spot detection algorithm and the cross-traffic detection algorithm is based on a gear status of the powertrain.

(7) According to a third aspect of the present disclosure, a vehicle includes a front, a rear, a driver side, and a passenger side. The vehicle further includes a first taillight assembly disposed at the rear on the driver side. The vehicle further includes a second taillight assembly disposed at the rear on the passenger side. The vehicle further includes a plurality of RADAR modules arranged at the rear of the vehicle and configured to detect an obstruction in a region exterior to the vehicle. The plurality of RADAR modules includes a first RADAR module in the first taillight assembly and having a first field of view. The plurality of RADAR modules includes a second RADAR module in the second taillight assembly and having a second field of view. The plurality of RADAR modules includes a third RADAR module disposed centrally between the first and second RADAR modules, the third RADAR module having a third field of view at least partially overlapping with each of the first and second fields of view behind the vehicle. The plurality of RADAR modules includes control circuitry in communication with the plurality of RADAR modules and configured to triangulate a position of the obstruction between a pair of the plurality of RADAR modules.

(8) These and other features, advantages, and objects of the present disclosure will be further understood and appreciated by those skilled in the art by reference to the following specification, claims, and appended drawings.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) In the drawings:

(2) FIG. 1 is a top-perspective view of a truck incorporating a detection system according to one aspect of the present disclosure;

(3) FIG. 2 is a block diagram of an exemplary detection system for use in a vehicle environment according to one aspect of the present disclosure;

(4) FIG. 3 is a rear perspective view illustrating geometric relationships between time-of-flight sensors employed at a rear of a vehicle and demonstrating fields of view of the time-of-flight sensors;

(5) FIG. 4 is a top view of a vehicle utilizing time-of-flight sensors to execute a blind spot detection algorithm initiated by control circuitry of the present disclosure;

(6) FIG. 5 is a top view of a vehicle utilizing time-of-flight sensors to execute a park assist algorithm initiated by control circuitry according to one aspect of the present disclosure;

(7) FIG. 6A is a rear perspective view of a bed of a truck housing cargo overhanging the bed of the truck;

(8) FIG. 6B is a top view of the vehicle of FIG. 6A, demonstrating the cargo having different overhanging lengths off of the bed of the truck;

(9) FIG. 6C is a side view of a tailgate in a closed position and an opened position in phantom;

(10) FIG. 7 is a top perspective view of a vehicle within a boundary box defined by control circuitry of a detection system according to one aspect of the present disclosure;

(11) FIGS. 8 and 9 are side perspective views of obstructions within available space above an extension of a vehicle;

(12) FIG. 10 is a perspective view of a vehicle incorporating a detection system configured to detect an inclining topography of a driveway;

(13) FIG. 11 is a side view of a vehicle incorporating a detection system configured to detect a declining topography of land;

(14) FIG. 12 is a perspective view of a towable device operably coupled with a vehicle incorporating a detection system configured to detect stability conditions of the towable device;

(15) FIG. 13 is a detailed perspective view of a hitch connection between a vehicle and a towable device and demonstrating fields of view of a detection system according to one aspect of the present disclosure;

(16) FIG. 14 is a rear perspective view of a dump trailer tilted upwardly at a pitch angle;

(17) FIG. 15 is a rear perspective view of a towable device operably coupled with a vehicle that incorporates a detection system configured to detect a length of the towable device; and

(18) FIGS. 16A-16C are side views of a towable device operably coupled with the vehicle and demonstrating a plurality of tilt angles of the towable device relative to a tailgate of the vehicle.

DETAILED DESCRIPTION OF PREFERRED EMBODIMENTS

(19) Reference will now be made in detail to the present preferred embodiments of the disclosure, examples of which are illustrated in the accompanying drawings. Wherever possible, the same reference numerals will be used throughout the drawings to refer to the same or like parts. In the drawings, the depicted structural elements may or may not be to scale and certain components may or may not be enlarged relative to the other components for purposes of emphasis and understanding.

(20) For purposes of description herein, the terms “upper,” “lower,” “right,” “left,” “rear,” “front,” “vertical,” “horizontal,” and derivatives thereof shall relate to the concepts as oriented in FIG. 1. However, it is to be understood that the concepts may assume various alternative orientations, except where expressly specified to the contrary. It is also to be understood that the specific devices and processes illustrated in the attached drawings, and described in the following specification are simply exemplary embodiments of the inventive concepts defined in the appended claims. Hence,

specific dimensions and other physical characteristics relating to the embodiments disclosed herein are not to be considered as limiting, unless the claims expressly state otherwise.

(21) The present illustrated embodiments reside primarily in combinations of method steps and apparatus components related to a detection system for a vehicle. Accordingly, the apparatus components and method steps have been represented, where appropriate, by conventional symbols in the drawings, showing only those specific details that are pertinent to understanding the embodiments of the present disclosure so as not to obscure the disclosure with details that will be readily apparent to those of ordinary skill in the art having the benefit of the description herein. Further, like numerals in the description and drawings represent like elements.

(22) As used herein, the term “and/or,” when used in a list of two or more items, means that any one of the listed items can be employed by itself, or any combination of two or more of the listed items, can be employed. For example, if a composition is described as containing components A, B, and/or C, the composition can contain A alone; B alone; C alone; A and B in combination; A and C in combination; B and C in combination; or A, B, and C in combination.

(23) As used herein, the term “about” means that amounts, sizes, formulations, parameters, and other quantities and characteristics are not and need not be exact, but may be approximate and/or larger or smaller, as desired, reflecting tolerances, conversion factors, rounding off, measurement error and the like, and other factors known to those of skill in the art. When the term “about” is used in describing a value or an end-point of a range, the disclosure should be understood to include the specific value or end-point referred to. Whether or not a numerical value or end-point of a range in the specification recites “about,” the numerical value or end-point of a range is intended to include two embodiments: one modified by “about,” and one not modified by “about.” It will be further understood that the end-points of each of the ranges are significant both in relation to the other end-point, and independently of the other end-point.

(24) The terms “substantial,” “substantially,” and variations thereof as used herein are intended to note that a described feature is equal or approximately equal to a value or description. For example, a “substantially planar” surface is intended to denote a surface that is planar or approximately planar. Moreover, “substantially” is intended to denote that two values are equal or approximately equal. In some embodiments, “substantially” may denote values within about 10% of each other, such as within about 5% of each other, or within about 2% of each other.

(25) As used herein the terms “the,” “a,” or “an,” mean “at least one,” and should not be limited to “only one” unless explicitly indicated to the contrary. Thus, for example, reference to “a component” includes embodiments having two or more such components unless the context clearly indicates otherwise.

(26) Referring generally to FIGS. **1-16C**, a detection system **10** is provided in connection with a vehicle **12** and is configured to detect objects, structures, terrain, or the like outside of and proximate to the vehicle **12**. The detection system **10** includes time-of-flight devices **14** that are arranged around the vehicle **12** to allow various detection techniques using a common set of sensors. For example, the time-of-flight devices **14** may be sensors that are configured to emit pulses of electro-magnetic waves at different frequencies depending on the operating mode. Accordingly, a dynamic, or multimode, system may be provided to allow enhanced detection techniques depending on a desired operation for the vehicle **12**.

(27) Reference to FIGS. **1-5**, according to a first example, the vehicle **12** includes a plurality of sides **16, 18, 20, 22** each having right and left portions. The plurality of sides **16, 18, 20, 22** includes a front **16**, a rear **18**, a driver side **20**, and the passenger side **22**. It is contemplated that the driver side **20** and passenger side **22** may be referred to left and right sides of the vehicle **12**, and that the vehicle **12** may be autonomous and/or may lack a driver or a passenger without deviating from the examples of the present disclosure. The time-of-flight devices **14** may include a plurality of radio detection and ranging (RADAR) modules **24, 26, 28** arranged at a common one of the sides **16, 18, 20, 22**, such as the rear **18** of the vehicle **12**, and configured to detect an obstruction

30 in a region exterior **32** to the vehicle **12**. The plurality of RADAR modules **24, 26, 28** includes a first RADAR module **24** on the driver side **20** of the vehicle **12**. The first RADAR module **24** has a first field of view **34**. A second RADAR module **26** is on the passenger side **22** of the vehicle **12** and has a second field of view **36**. A third RADAR module **28** is spaced from the first RADAR module **24** at a first distance **38** and the second RADAR module **26** at a second distance **40**. The third RADAR module **28** has a third field of view **42** at least partially overlapping with each of the first and second fields of view **34, 36** behind the vehicle **12**. Control circuitry **44** is in communication with the plurality of RADAR modules **24, 26, 28** and is configured to triangulate a position of the obstruction **30** between a first pair of the plurality of RADAR modules **24, 26, 28**. The first distance **38** can be substantially equal to the second distance **40**.

(28) In some examples, the third RADAR module **28** is disposed between the first and second RADAR modules **24, 26**. The third RADAR module **28** may be disposed centrally between the first and second RADAR modules **24, 26**. For example, the first distance **38** may be between a quarter of (25%) and four times (400%) of the second distance **40**. The first distance **38** may be between a half of (50%) and twice (200%) of the second distance **40**. The third RADAR module **28** may be vertically spaced from the first and second RADAR modules **24, 26**. The first RADAR module **24** may be vertically aligned with the second RADAR module **26**. In further examples, the vehicle **12** includes a first taillight assembly **46** disposed at the rear **18** on the driver side **20**, as well as a second taillight assembly **48** disposed at the rear **18** on the passenger side **22**. The first and second RADAR modules **24, 26** are disposed in the first and second taillight assemblies **46, 48**, respectively.

(29) In some examples, the control circuitry **44** is configured to selectively execute a blind spot detection algorithm to determine an object **50** in a blind spot **52** of at least one of the driver side **20** and the passenger side **22** based on the positional data captured in the first and second fields of view **34, 36**. The control circuitry **44** may also, or alternatively, selectively execute a cross-traffic detection algorithm to determine an object **50** is a cross-traffic area using the first and second fields of view **34, 36**. For example, regions behind and to the right or left may be the cross-traffic areas where other vehicles may be detected during reversing of the vehicle **12**. The vehicle **12** may further include a powertrain **54** in communication with the control circuitry **44**. Selective execution of the blind spot **52** detection algorithm and the cross-traffic detection algorithm may be based on a gear status of the powertrain. For example, the control circuitry **44** may be configured to execute the blind spot detection algorithm and the cross-traffic detection algorithm when the gear status is not park. The control circuitry **44** is configured to execute a parking assist algorithm that guides steering of the vehicle **12** based on the positional data captured in both the third field of view **42** and at least one of the first and second fields of view **34, 36**. The control circuitry **44** may be configured to select between the park assist algorithm and the blind spot **52** detection algorithm based on the gear status. For example, if the vehicle **12** is in reverse gear, the park assist mode may be activated.

(30) In some examples, the vehicle **12** further includes a tailgate **56** at the rear **18** of the vehicle **12**. The third RADAR module **28** may be operably coupled with the tailgate **56**. The control circuitry **44** may further be configured to execute a gesture recognition algorithm that determines a gesture of a user behind the vehicle **12** based on the positional information captured in the third field of view **42**. The control circuitry **44** may further be configured to execute the gesture recognition algorithm when the gear status is park.

(31) In some examples, the control circuitry **44** is further configured to triangulate the position of the obstruction **30** between a second pair of the plurality of RADAR modules **24, 26, 28**, compare the triangulation based on the second pair of the plurality of RADAR modules **24, 26, 28** to the triangulation based on the first pair of the plurality RADAR modules, and update the position based on the comparison.

(32) Referring to FIGS. **1-6C**, in a second example, the detection system **10** for the vehicle **12**

includes the at least one time-of-flight device **14** configured to capture the positional information about cargo **58** in the vehicle **12** and the obstruction **30** outside the vehicle **12**. The detection system **10** further includes control circuitry **44** that is in communication with the at least one time-of-flight device **14**. The control circuitry **44** is configured to determine a location of the obstruction **30** relative to a rear **18** of the vehicle **12**. The control circuitry **44** is further configured to calculate a length **60** of at least one overhang **62, 64** of the cargo **58** based on the positional information, estimate a lateral position **66** of the at least one overhang **62, 64** based on the positional information, determine at least one endpoint **68, 70** of the cargo **58** based on the lateral position **66** and the length **60**, determine at least one potential contact point **72, 74** between the cargo **58** and the obstruction **30** based on the location and the at least one endpoint **68, 70**, and generate an output based on the at least one potential contact point **72, 74**. The detection system **10** of the present example may provide for enhanced availability when adjusting a position of the vehicle **12** by identifying additional space for movement of the vehicle **12** while limiting potential contact.

(33) In some examples, the cargo **58** includes a first object **78** and a second object **80**. The control circuitry **44** is further configured to determine a first endpoint **68** for the first object **78** based on the length **60** and the lateral position **66** of a first overhang **62** of the first object **78**. The control circuitry **44** is further configured to determine a second endpoint **70** for the second object **80** based on the length **60** and a lateral position **66** of a second overhang **64** of the second object **80**. The control circuitry **44** is further configured to determine a first potential contact point **72** between the first object **78** and the obstruction **30** based on the first endpoint **68**. The control circuitry **44** is further configured to determine a second potential contact point **74** between the second object **80** and the obstruction **30** based on the second endpoint **70**.

(34) In some examples, the control circuitry **44** is further configured to compare the first potential contact point **72** with the second potential contact point **74** and calculate a steering profile **82** based on the comparison of the first potential contact point **72** with the second potential contact point **74**. The detection system **10** may further include a steering system **84** that adjusts a steering angle of the vehicle **12** in response to the steering profile **82**. A display **88** may be in communication with the control circuitry **44** and configured to present an indication of the steering profile **82** for the steering system **84** in response to the output. The indication may include an adjustment for the steering angle to align with the steering profile **82**.

(35) In some examples, the control circuitry **44** is further configured to determine a clearance zone between the first and second potential contact points **72, 74** based on the location of the obstruction **30**. The detection system **10** may further include a brake system **92** that adjusts a speed of the vehicle **12** in response to the output. The control circuitry **44** may further be configured to determine a proximity of the cargo **58** to at least one of the first potential contact point **72** and the second potential contact point **74** and communicate an instruction to the brake system **92** to adjust the speed of the vehicle **12** in response to the proximity.

(36) In some examples, the display **88** is configured to present the at least one potential contact point **72, 74** in response to the output. The at least one time-of-flight device **14** may include the plurality of RADAR modules **24, 26, 28** disposed along the rear **18** of the vehicle **12** and configured to detect the obstruction **30** behind the vehicle **12**. The plurality of RADAR modules **24, 26, 28** includes the first RADAR module **24** aligned with the driver side **20** of the rear **18** and the second RADAR module **26** aligned with the passenger side **22** of the rear **18**. The plurality of RADAR modules **24, 26, 28** includes the third RADAR module **28** disposed between the first and second RADAR modules **24, 26**. The third RADAR module **28** may be disposed above or below each of the first and second RADAR modules **24, 26**.

(37) Referring to FIGS. 1-5 and 7-11, in a third example, the vehicle **12** includes a passenger compartment **94** having a side, such as a sidewall **96** or outer wall of the passenger compartment **94**, and an upper wall **98** that extends from the side in an upper plane. An extension **100** extends from the side of the passenger compartment **94** to an end **102** that extends in an end plane. The at

least one time-of-flight device **14** is configured to capture positional information about the obstruction **30** outside the vehicle **12**. The control circuitry **44** is in communication with the at least one time-of-flight device **14** and is configured to define a space above the extension **100** between the side, the upper plane, and the end plane. For example, the space may be one of a plurality of voids **106** around the vehicle mapped by the control circuitry **44**. The control circuitry **44** is further configured to calculate, based on the positional information, an available position for the vehicle **12** having the obstruction **30** in the space. The control circuitry **44** is further configured to generate an output in response to the available position. This arrangement may provide for enhanced spatial utilization when maneuvering the vehicle **12**.

(38) In some examples, the vehicle **12** further includes the powertrain **54** that moves the vehicle **12** in a motion direction. The control circuitry **44** may further be configured to communicate an instruction to adjust the powertrain in response to the calculation of the available position. The control circuitry **44** may further be configured to determine a first potential contact point **72** between the obstruction **30** and the end **102**, determine a second potential contact point **74** between the obstruction **30** and the sidewall **96**, and compare the first and second potential contact points **72**, **74** to calculate the available position. The instruction to adjust the powertrain may be based on the comparison.

(39) In some examples, the control circuitry **44** is further configured to calculate a first contact distance **112** between the first potential contact point **72** and the end **102**. The control circuitry **44** is further configured to calculate a second contact distance **114** between the second potential contact point **74** and the sidewall **96**. The control circuitry **44** is further configured to assign a maximum travel for the vehicle **12** in the motion direction to one of the first contact distance **112** and the second contact distance **114**. The control circuitry **44** may further be configured to assign the maximum travel based on the shorter of the first contact distance **112** and the second contact distance **114**.

(40) In some examples, the vehicle **12** includes the brake system **92** that adjusts the speed of the vehicle **12** in response to the maximum travel. The extension **100** may include an outer compartment **118** of the vehicle **12**. For example, the outer compartment **118** may be a trunk or a frunk of the vehicle **12**. The space may be defined above a trunk **120** or a hood **122** of the vehicle **12**, though other spaces (e.g., spaces under the chassis, above the hood, or any other spaces around the vehicle **12**) may be defined between a boundary box **196** (FIG. 7) around the vehicle **12** and the vehicle **12** itself.

(41) In some examples, the outer compartment **118** is a bed **124** of a truck **126**. The vehicle **12** may further include a cover **128** that is selectively disposed in the space for covering the bed **124** of the truck **126**. For example, the cover **128** may be a cap for the vehicle **12** that selectively covers the outer compartment **118** (e.g., a detachable truck cap). The control circuitry **44** may be configured to determine the presence of the cover **128**. An input may be in communication with the control circuitry **44** and configured to detect the cover **128**. For example, the input may be a signal generated by a detection sensor that monitors the presence of the cover **128**. The vehicle **12** may further include the steering system **84** that adjusts the steering angle of the vehicle **12** to align the vehicle **12** with the available position.

(42) Referring to FIGS. 1-5 and 12-16C, in a fourth example, the detection system **10** for the vehicle **12** includes the at least one time-of-flight device **14** that is configured to capture positional information about a contour of a towable device **130** outside the vehicle **12**. The detection system **10** further includes a user interface configured to present messages to a user. The control circuitry **44** is in communication with the at least one time-of-flight device **14** and the user interface. The control circuitry **44** is configured to calculate a tilt **131** of the contour based on the positional information about the contour, compare the tilt **131** to a threshold angle **132** for the contour, determine at least one stability condition of the towable device **130** based on the comparison of the tilt **131** to the threshold angle **132**, and communicate an instruction to the user interface to indicate

the at least one stability condition. The detection system **10** may provide for detecting unstable conditions or potentially unstable conditions of the towable device **130** by employing the enhanced arrangement of time-of-flight sensors, according to some aspects of the present disclosure.

(43) It is contemplated that the towable device **130** may embody any apparatus or device configured to be coupled with (e.g., hitched) to the vehicle **12** to allow the vehicle **12** to tow the towable device **130**. For example, the towable device may be another vehicle, such as a recreational vehicle (RV) or an automotive vehicle (e.g., car, truck), a trailer, a camper, or any other towable device **130**.

(44) According to some examples, the towable device **130** is operably coupled with the vehicle **12**. The contour may include a front wall **138** of the towable device **130**, and the control circuitry **44** may further be configured to calculate a pitch of a floor **142** of the towable device **130** based on the tilt **131** of the front wall **138**. For example, the pitch may be in angular deviation from a driving surface **144** under the vehicle **12** in an inclined or declined position. The control circuitry **44** may further be configured to determine a topography **146** of an area behind the vehicle **12** based on the pitch. For example, the control circuitry **44** may determine the presence of an incline or decline behind the vehicle **12**. The towable device **130** may be a dump trailer **150** having the floor **142** pivotable relative to a frame **152** of the towable device **130**. A lift unit **153** may be in communication with the control circuitry **44** and configured to adjust the pitch of the floor **142** of the dump trailer **150**.

(45) According to some examples, the at least one stability condition includes an unstable condition corresponding to the pitch exceeding the threshold angle **132**. The control circuitry **44** may further be configured to communicate an instruction to interrupt the lift unit **153** adjusting the pitch in response to the unstable condition. The control circuitry **44** may further be configured to estimate a length of the towable device **130** based on the positional information and calculate the threshold angle **132** based on the length of the towable device **130**. The detection system **10** may further include an input indicating a loaded state or an unloaded state of the towable device **130**. The control circuitry **44** may be configured to adjust threshold angle **132** based on the input. The input may be a signal from one of a selection at the user interface and a weight sensor **154** for the towable device **130**. For example, the weight sensor **154** may be operably coupled with the floor **142** of the towable device **130** and configured to provide weight information wirelessly or over a wired connection to the control circuitry **44** to indicate a weight of the towable device **130**.

(46) The at least one time-of-flight device **14** may include the plurality of RADAR modules **24**, **26**, **28** arranged near the tailgate **56** of the vehicle **12**. A pair of the plurality of RADAR modules **24**, **26**, **28** are disposed in the taillight assemblies **46**, **48** of the vehicle **12**. In some examples, at least one of the plurality of RADAR modules **24**, **26**, **28** is disposed in another lighting assembly, such as a headlight assembly **158**, **160** of other lighting assembly that illuminates the region exterior **32**.

(47) Referring now to FIG. **1**, the vehicle **12** may be a truck **126** having the bed **124** selectively accessible via the tailgate **56**. For example, the tailgate **56** may have a handle **156** for selectively opening or closing the tailgate **56**. The third RADAR module **28** may be disposed adjacent to the handle **156** for the tailgate **56** and vertically offset from the first and second RADAR modules **24**, **26** in the first and second taillight assemblies **46**, **48**, respectively. Although illustrated as a truck **126**, it is contemplated that the vehicle **12** may be any automotive vehicle, such as a car, a van, a sedan, a sport utility vehicle, or any other automotive vehicle. Further, it is contemplated that the plurality of RADAR modules **24**, **26**, **28** may be positioned additionally, or in an alternative, at the front **16** of the vehicle **12**, such as having the first and second RADAR modules **24**, **26** disposed in first and second headlight assemblies **158**, **160** of the vehicle **12**. In this example, the third RADAR module **28** may be disposed between the first and second RADAR modules **24**, **26** at, above, or below, the first and second RADAR modules **24**, **26**. In general, a triangular arrangement of the plurality of RADAR modules **24**, **26**, **28** may be provided along the front **16** or rear **18** of the vehicle for monitoring areas in front of or behind the vehicle **12**.

(48) The time-of-flight devices **14** described herein may refer to the RADAR modules **24, 26, 28** as described, or may refer to any other time-of-flight device **14**, such as a light detection and ranging unit (LiDAR), a Sound Navigation and Ranging (Sonar) unit, or any other time-of-flight device **14**. In general, the arrangement of the time-of-flight devices **14** relative to the motion direction of the vehicle **12** may provide for the enhanced range of detection and locating of the present detection system **10**. The RADAR may have a range of between 70 and 90 gigahertz (GHz), though other RADAR ranges may be employed. For example, the RADAR may be between 76 GHz and 81 GHz.

(49) Referring now to FIG. 2, an exemplary functional diagram of the control circuitry **44** of the detection system **10** of the present disclosure is shown. The control circuitry **44** may include a controller **162** having a processor **164** and a memory **166** that stores instructions. The processor **164** may execute the instructions to cause the controller **162** to operate the time-of-flight devices **14** and control one or more of the brake system **92**, the powertrain **54**, the lift unit **153**, or another vehicle system in response to the positional data collected by the plurality of RADAR modules **24, 26, 28**. In some examples, the control circuitry **44** may communicate signals to adjust the frequencies of the radio waves emitted by the RADAR modules **24, 26, 28** and/or an orientation of the emissions of radio waves to focus on particular regions around the vehicle **12**. In other examples, the RADAR modules **24, 26, 28** operate in a constant detection mode and the processor **164** operates in variable algorithms depending on the particular operation performed by the detection system **10**. For example, the control circuitry **44** may be configured to operate in the blind spot detection mode, the park assist mode, or a backup assist mode. Depending on the particular mode selected, the processor **164** may execute different instructions that causes the processor **164** to read segments of the data collected by the plurality of RADAR modules **24, 26, 28**. Stated differently, the different operating modes for the processor **164** may be associated with different locations around the vehicle **12** to be monitored and, as a result, the positional data from each of the time-of-flight devices **14** may be read at varying rates or sampled at different frequencies in order to analyze the positional data pertinent to that particular location. In this way, computational power and latency may be enhanced by the detection system **10**.

(50) In some examples, the control circuitry **44** is configured to compare the positional data captured by the plurality of RADAR modules **24, 26, 28** in order to triangulate the positions of the objects **50, 78, 80** or obstructions **30** around the vehicle **12**. For example, the time-of-flight of radio waves emitted and received from the first RADAR modules **24** may be compared to the time-of-flight of radio waves emitted received by the second RADAR module **26** in order to triangulate the position of the obstruction **30**.

(51) The detection system **10** further includes a wireless network **168** that may provide for communication between the control circuitry **44** and one or more mobile devices **170**. For example, the wireless network **168** may be operable with Wi-Fi®, Bluetooth®, ZigBee®, SMS, or any other wireless communication protocol, such as any short- or long-wave communications. The mobile device **170** may incorporate ultra-wideband (UWB) RADAR functionality and serve as one of the plurality of RADAR modules **24, 26, 28** (e.g., a fourth RADAR module). For example, a user may be positioned near the tailgate **56** and may use the mobile device **170** to guide a driver of the vehicle **12** or guide steering of the vehicle **12** and a backing up operation by employing the RADAR module in the mobile device **170** to gather additional positional data about obstructions **30** or objects **50, 78, 80** behind the vehicle **12**. In addition, or in an alternative, the RADAR modules **24, 26, 28** on the vehicle **12** may detect the presence of the user near the tailgate **56** and control opening or closing of the tailgate **56** based on the location of the user. For example, the detection system **10** may include an electronic door lock **172** and/or an actuator **174** configured to control opening or closing of the tailgate **56**. Based on the detection of the user within an opening path of the tailgate **56**, the control circuitry **44** may limit movement of the tailgate **56** when the user, or another obstruction **30**, is within the opening movement. When the tailgate **56** is in the open

position, a pair of the plurality of RADAR modules **24, 26, 28** may be configured to detect the presence of the user and limit a closing movement of the tailgate **56** based on the user being within a region near the tailgate **56**.

(52) In some examples, the at least one time-of-flight device **14** includes UWB transceivers **171**, or beacons/antennas, positioned at the rear **18** of the vehicle **12**. For example, the UWB transceivers **171** may be incorporated in rear corners and front corners of the vehicle **12** to provide 360° of RADAR coverage around the vehicle **12**. The UWB transceivers **171** may provide for enhanced locating functionality to detect precise location of the objects **50, 78, 80** proximate to the vehicle **12**.

(53) With continued reference to FIG. 2, the powertrain **54** may include the steering system **84**, an engine **175a** of the vehicle **12**, a transmission system **175b** of the vehicle **12**, or any other system configured to control movement of the vehicle **12**. As will be described further herein, the control circuitry **44** may be configured to adjust steering of the vehicle **12** in response to the positional information captured by the RADAR modules **24, 26, 28**. In some examples, the gear state of the transmission may be accessed by the control circuitry **44** and be employed to determine the operational mode for the processor **164**. In other examples that will be described in reference to FIGS. 12 through 16C, the control circuitry **44** may be in communication with the lift unit **153** of the dump trailer **150** that may be operably coupled with the vehicle **12**. The lift unit **153** may incorporate any electromechanical actuating device such as a motor **176** or a hydraulic unit that may drive the floor **142** of the tractor upwardly and downwardly, as will be described further herein. Further, the weight sensor **154** may be in communication with the control circuitry **44** to allow the control circuitry **44** to determine various states of the towable device **130** as will be further described herein.

(54) Still referring to FIG. 2, a human-machine interface (HMI) may be provided in the vehicle **12** to allow the user to interact with various aspects of the detection system **10**. For example, the HMI may incorporate the display **88** and may serve as the interface **177** previously described. In some examples, the mobile device **170** serves as the interface **177** and the display **88**. For example, the steering profile **82**, the stability conditions for the towable device **130**, the presence and location of objects **50, 78, 80** or obstructions **30**, and available parking positions or other movements of the vehicle **12** may be presented at the display **88** in the form of an image, text, numbers, or any other indication of detection by the detection system **10**.

(55) Referring now more particularly to FIG. 3, a geometric arrangement **178** for the three RADAR modules **24, 26, 28** is illustrated in the tailgate **56** of the vehicle **12**. More particularly, the first and second RADAR modules **24, 26** are incorporated into the first and second taillight assemblies **46, 48**, respectively, and the third RADAR module **28** is incorporated centrally within the tailgate **56**. In this exemplary arrangement **178**, the first distance **38** is substantially equal to the second distance **40**, and the third RADAR module **28** is vertically, or vehicle-upwardly, offset from the first and second RADAR modules **24, 26** which are in a common horizontal plane. Thus, the arrangement **178** of the plurality of RADAR modules **24, 26, 28** forms an isosceles triangular arrangement **178** to provide for enhanced scanning and detection equally between the driver side **20** or the passenger side **22** behind the vehicle **12**. Further, by providing a vertical offset between at least a pair of the RADAR modules **24, 26, 28**, an expansive field of view amongst the three fields of view **34, 36, 42** may be provided. For example, the third field of view **42** may overlap with the first and second fields of view **34, 36** in a smaller region than if all of the RADAR modules **24, 26, 28** were vertically aligned with one another.

(56) Referring generally to FIGS. 2 and 3, control circuitry **44** may further be configured to execute the gesture recognition algorithm and utilize the positional information captured from the RADAR modules **24, 26, 28**. For example, when the user is near the rear **18** of the vehicle **12**, the position of the user may be identified via the RADAR modules **24, 26, 28** and various gestures of the user may be detected based on the positional information gathered by the RADAR modules **24, 26, 28**. For

example, raising of the foot, moving of the hands, or other movements of the body of the user may be detected based on the positional information in the fields of view **34, 36, 42** and cause opening or closing of the tailgate **56**. As previously described, the opening or closing of the tailgate **56** may be limited based on the location of the user. For example, various gestures to open or close the tailgate **56** may be detected from close or far distances from the vehicle **12** within the fields of view **34, 36, 42** of the RADAR modules **24, 26, 28**. In some examples, the control circuitry **44** executes the gesture recognition algorithm when the gear state is park and the engine is on. Continuing with this example, when the vehicle **12** is in reverse, the park assist operating mode may be activated. The park assist mode may include the cross-traffic detection algorithm. For example, the park assist mode may cause the first and second RADAR modules **24, 26** to scan the cross traffic areas when the gear status is reverse. When the gear status is drive, the control circuitry **44** may execute the blind spot **52** detection algorithm. By providing the redundancy of the plurality RADAR modules on the vehicle **12**, multi-mode functionality for the detection system **10** may be provided.

(57) The arrangement **178** of the RADAR modules **24, 26, 28** may further provide for enhanced redundancy in the event of limited functionality of one of the three RADAR modules **24, 26, 28**. Further, by providing the arrangement of the plurality of RADAR modules **24, 26, 28**, ultrasonic sensors typically incorporated in bumpers or other exterior portions of the vehicle **12** may be omitted to provide enhanced efficiency in manufacturing. Accordingly, cutouts for ultrasonic sensors and multiple wired connections may be reduced.

(58) Referring now to FIGS. **4** and **5**, examples of the blind spot detection algorithm (FIG. **4**) and the park assist algorithm (FIG. **5**) are illustrated. As shown, the first and second RADAR modules **24, 26** incorporated into the taillight assemblies **46, 48** may allow for dual functionality of the RADAR modules **24, 26, 28** for various operations of moving the vehicle **12**. As previously described, incorporation of the third RADAR module **28** may allow for greater redundancy in the event of limited functionality of at least one of the RADAR modules **24, 26, 28** to allow for the blind spot detection algorithm and the park assist algorithm to be executed with two RADAR modules **24, 26**.

(59) Referring now to FIGS. **6A** and **6B**, the plurality of RADAR modules **24, 26, 28** may be employed for detection of extended loads within the vehicle **12**. For example, the cargo **58** in the bed **124** of the truck **126** may extend beyond the rear **18** of the truck **126**. In such an example, the effective length **180** of the vehicle **12** may be the length of the vehicle **12** plus the overhang **62, 64** of the cargo **58**. Accordingly, by detecting the at least one endpoint **68, 70** of the at least one overhang **62, 64** using the positional information from the RADAR sensors, the present detection system **10** may allow the control circuitry **44** to provide an enhanced estimate for maneuvering of the vehicle **12**. For example, changing lanes of the vehicle **12** without interaction with or contact with another vehicle or another item may be limited by providing these detection techniques. In addition, or in an alternative, the detection system **10** may enhance the blind spot detection algorithm by incorporating the length **60** of the at least one overhang **62, 64** into estimation of available maneuvering of the vehicle **12**. Accordingly, the detection system **10** may provide for enhanced functionality to detect and characterize lengths of extended loads on the vehicle **12**.

(60) Still referring to FIGS. **6A** and **6B**, the three fields of view **34, 36, 42** of the three RADAR modules **24, 26, 28** may be oriented for accessibility to the tailgate **56** and for blind spot **52** identification as described previously. In this example, the control circuitry **44** may modify operation of the third RADAR module **28** to monitor positional information captured from a direction that would ordinarily be upward when the tailgate **56** is in the closed position. For example, the control circuitry **44** may operate a tailgate-open algorithm and a tailgate-closed algorithm in which, depending on the algorithm executed, positional data captured from an upward angle of the radio waves are monitored or radio waves captured in a downward orientation are monitored. For example, with reference to FIG. **6C**, in the closed position of the tailgate **56** (e.g., the tailgate **56** is up), the control circuitry **44** may actively read data captured in a first angular

range **182** (e.g., between 45° and 60° from center) and ignore data captured in a second angular range **184** (e.g., between 1° and 45° outside of the first angular range **182**). However, when the tailgate **56** is the open position (e.g., the tailgate **56** is down), the control circuitry **44** may actively read positional information captured in the second angular range **184** and ignore positional information captured in the first angular range **182**. Accordingly, detection of the tailgate **56** position may be determined based on the positional information generated by the first and second RADAR modules **24**, **26**. In other examples, various sensors, such as the weight sensor **154**, a door open/closed sensor **188**, or any other detection sensor may be employed by the detection system **10** and in communication with the control circuitry **44** to determine the position of the tailgate **56**.

(61) Referring back to FIGS. **6A** and **6B**, when the extended load is present and detected by the RADAR modules **24**, **26**, **28**, each of the RADAR modules **24**, **26**, **28** may be focused or oriented toward detection of the length **60** of the at least one overhang **62**, **64**. With particular reference to FIG. **6B**, gaps **194** may be identified within the at least one overhang **62**, **64** that may allow for enhanced spatial optimization for maneuvering of the vehicle **12**. For example, a reversing operation of the vehicle **12** is shown with a steering profile **82** for the vehicle **12** between an initial position **190** and an intermediate position **192**. In the example in which the cargo **58** includes a plurality of objects (e.g., the first object **78** and the second object **80**), each object **78**, **80** has an overhang **62**, **64** of the length **60**. Depending on the lateral position **66** (e.g., the passenger side **22** to driver side **20** direction) the obstruction **30** may be limited from contact with the vehicle **12** or the cargo **58** while allowing the vehicle **12** to move to a target location near the instruction by utilizing the gaps **194** identified by the control circuitry **44**. For example, the control circuitry **44** may determine the first potential contact point **72** between the first endpoint **68** and the obstruction **30** and the second potential contact point **74** between the second endpoint **70** and the obstruction **30**. The potential contact points **72**, **74** may be determined utilizing the lateral position **66** as calculated based on the positional information gathered by the plurality of RADAR modules **24**, **26**, **28**. Based on these potential contact points **72**, **74**, the control circuitry **44** may communicate instructions to the steering system **84** to adjust the steering angle of the vehicle **12** to align with the steering profile **82** for the maneuvering of the vehicle **12**.

(62) As described, the gaps **194** identified based on the positional information may serve as a clearance zone between the potential contact points **72**, **74**. For example, as demonstrated in FIG. **6B**, the first object **78** and the second object **80** have different lengths **60** of overhang **62**, **64** in the vehicle-rearward direction, and therefore, the clearance zone is on the driver side **20** of the vehicle **12** for the obstruction **30**. Accordingly, in this example, the vehicle **12** may back up to a position in which the obstruction **30** is in the clearance zone. The control circuitry **44** may communicate instructions to adjust the steering to provide for this position. In an alternative, the control circuitry **44** may communicate instructions to indicate to the driver the clearance zone and the available positions for maneuvering the vehicle **12**.

(63) In addition to the parking assistance features provided by the detection system **10** as described, cross-traffic alert operations performed by the control circuitry **44** and/or the blind spot **52** detection algorithms executed by the control circuitry **44** may be enhanced by utilizing the positional information captured by the three RADAR modules **24**, **26**, **28**. Accordingly, the clearance zones determined by the control circuitry **44** may allow for enhanced maneuverability of the vehicle **12** in various scenarios.

(64) As previously described with respect to FIG. **2**, the display **88**, such as the display **88** of the human-machine interface **177** and/or the mobile device **170**, may be configured to present an indication of the clearance zone and/or the steering profile **82** to guide the user in controlling the vehicle **12** to limit contact with the obstruction **30**. Further, the brake system **92** may be controlled via instructions from the control circuitry **44** in response to proximity to the obstruction **30**. In this way, the vehicle **12** may be guided to limit contact with the instruction using enhanced positional information, such as lengths **60** of the overhangs **62**, **64**.

(65) In other nonlimiting examples, the plurality of RADAR modules **24, 26, 28** may be employed to detect a water surface and a water level when backing up the vehicle **12** or pulling a marine vessel at the entry of a body of water. For example, as will be described in reference to FIGS. **12-16C**, if the marine vessel is present on the towable device **130**, the first and second RADAR modules **24, 26** may detect the length of the towable device **130** and therefore the length of the marine vessel to allow the control circuitry **44** to determine whether the marine vessel is in the body of water. In this way, the arrangement **178** of the RADAR modules **24, 26, 28** may limit the vehicle **12** from entering the body of water. Accordingly, the control circuitry **44** may communicate with the powertrain **54** and/or the brake system **92** to indicate to the user or to automatically adjust the powertrain and/or the brake system **92** based on estimation of how far the marine vessel is in the body of water.

(66) Referring now to FIG. **7-11**, the detection system **10** may be employed for enhanced spatial availability when maneuvering a vehicle **12**, such as during a parking operation for the vehicle **12**, reversing the vehicle **12**, or advancing the vehicle **12** when a driving surface **144** of the region exterior **32** to the vehicle **12** or under the vehicle **12** is uneven, slanted, or has unexpected features. Further, the detection system **10** may provide for enhanced spatial optimization for obstructions **30** above the vehicle **12**, below the vehicle **12**, or otherwise near the vehicle **12** that may be disposed in voids **106** within a boundary box **196** of the vehicle **12**.

(67) Referring more particularly to FIG. **7**, a rectangular boundary box **196** encompassing the vehicle **12** may be defined by the control circuitry **44** of the detection system **10**. The boundary box **196** may be a representative of space occupied by the vehicle **12** and free space, or availability, for the obstruction **30** to be disposed in. For example, void **106** above the bed **124** of the truck **126** or above the hood **122** of the front **16** compartment of the truck **126** may be available for obstructions **30**. Accordingly, and consistent with previous examples of enhancing availability of space for the vehicle **12**, the detection system **10** may allow for enhanced ease-of-use and maneuverability for the vehicle **12** during a reversing operation or parking operation.

(68) Still referring to FIG. **7**, the upper wall **98** of the passenger compartment **94** may extend along an upper plane. For example, the upper plane may extend vertically. The end **102** of the vehicle **12**, which may be the rear **18** or the front **16** of the vehicle **12**, may also extend in a plane, such as an end plane that also extends vertically and parallel to the upper plane. The sides of the vehicle **12** may also generally or substantially extend in a vertical plane. Thus, voids **106** may be formed within the boundary box **196** in which the obstruction may be disposed. For example, the control circuitry **44** may define the void **106** above the extension **100** (e.g., the bed **124** of the truck **126**) between the upper plane and the end plane. For example, available space may be above the bed **124** of the truck **126** and/or above the hood **122** of the truck **126** between a front of the passenger compartment **94** of the vehicle **12**. Other voids **106** may be below a chassis **200** of the vehicle **12** to allow for obstructions **30** to be disposed below the chassis **200** without interfering with maneuvering of the vehicle **12**. For example, bumps or uneven terrain may be scanned by the RADAR modules **24, 26, 28** and determined, by the control circuitry **44**, to be disposed in the available space under the vehicle **12**. Thus, the arrangement **178** of the RADAR modules **24, 26, 28** in the tailgate **56** and the taillights may allow for enhanced maneuvering of the vehicle **12** on uneven terrain and/or about obstacles at a height of the passenger compartment **94**.

(69) Still referring to FIG. **7**, the boundary box **196** may generally be bounded by vertical planes extending along the front **16**, the rear **18**, and the sides of the vehicle **12**, as well as horizontal planes extending at a topmost point of the vehicle **12** and a bottommost point of the vehicle **12** (e.g., contact of the tires **202** of the vehicle **12** with the driving surface **144**). Thus, the voids **106** described herein may be any crevice or void accessible from outside of the boundary box **196** by the obstructions **30** which may protrude into the voids **106**.

(70) Referring now to FIGS. **8-11**, exemplary utilization of the voids **106** are generally demonstrated. With reference to FIG. **8** in particular, a parallel parking operation may be enhanced

by allowing the vehicle **12** (e.g., a car) to be parked behind an obstruction **30** that may protrude into of the voids **106** above the hood **122** of a front compartment of the vehicle **12**. In this example, the three RADAR modules **24, 26, 28** may be incorporated into the two headlights **158, 160** and a central location between the two headlights **158, 160**. As shown, the first potential contact point **72** between the obstruction **30** and the end **102** may be determined by the control circuitry **44**, and the second potential contact point **74** between the obstruction **30** and the upper plane may be determined by the control circuitry **44**. In this example, the upper plane may extend at an oblique angle with a windshield **204** of the vehicle **12** to form an oblong-shaped voids **106**. The control circuitry **44** may compare the first and second potential contact points **72, 74** to calculate the available position for the obstruction **30**. As depicted, the first contact distance **112** between the first potential contact point **72** and the end **102** (e.g., the front **16**), may be calculated and compared to the second contact distance **114** between the second potential contact point **74** and the upper wall **98**. Depending on the magnitude of the first and second contact distances **112, 114**, the maximum travel for the vehicle **12** in the motion direction (e.g., vehicle-forward) may be assigned by the control circuitry **44**. For example, the shorter of the first contact distance **112** and the second contact distance **114** may be assigned to the maximum travel for the vehicle **12** to limit contact with the obstruction **30**. It is contemplated that the first and second contact distances **112, 114** may be aligned with the motion direction. For example, as will be described in reference to FIG. 9, the motion direction may be vehicle-rearward.

(71) In some examples, the voids **106** may be excluded for utilization by the control circuitry **44**. For example, if a void **106** above the front compartment of the vehicle **12** is mapped by the control circuitry **44** using RADAR, the control circuitry **44** may determine that this void **106** is not free for receiving the external obstruction **30**. Thus, height detection by the present time-of-flight devices **14** may limit contact with the obstruction **30** by detecting the forward-most point of the vehicle **12** and the rearward-most point of the obstruction **30** (or vice versa). The control circuitry **44** may then compare these points and determine an available position for the vehicle **12** based on the two points not crossing a common vertical plane **205**. In other words, the void **106** may not be utilized for the obstruction **30**, but rather be free space from the obstruction **30**. Thus, in the example illustrated in FIG. 8, the control circuitry **44** may limit the vehicle **12** from automatic or manual maneuvering into the position illustrated and instead allow the vehicle **12** to move behind the vertical plane **205**.

(72) With reference particularly to FIG. 9, the obstruction **30** may be above the rear **18** of the vehicle **12**, and the controller **162** may define the available space above the rear **18** of the vehicle **12**. In this example, multiple contact points may be determined by the control circuitry **44** due to the obstruction **30** having an irregular shape to allow a part of the obstruction **30** is disposed within the available space while another part of the obstruction **30** to the outside of the boundary box **196**. In this example, the first contact distance **112** is greater than the second contact distance **114**. Accordingly, the control circuitry **44** may communicate an instruction to stop reversing of the vehicle **12** as the second contact distance **114** approaches zero. Accordingly, the available position for the vehicle **12** may have at least a portion of the obstruction **30** disposed in the available space above the compartment.

(73) Referring now to FIGS. 10 and 11, variations in terrain or the topography of the driving surface **144** may present obstructions **30** for available spaces underneath the chassis **200** of the vehicle **12**. For example, by utilizing the plurality of RADAR modules **24, 26, 28** in the rear **18** or in the front **16** of the vehicle **12**, obstructions **30** presented by the driving surface **144** may be detected and allow the detection system **10** to limit contact of the vehicle **12** with these obstructions **30**. For example, curbs along a curved pathway (e.g., a driveway) may be detected by the RADAR modules **24, 26, 28** despite extending alongside the vehicle **12**. In addition, weather conditions, such as snow and ice accumulation may be detected by using the first, second, and third RADAR modules **24, 26, 28** around the region exterior **32** to the vehicle **12**. Based on the positional information captured by the RADAR modules **24, 26, 28**, the control circuitry **44** may

communicate instructions to adjust the steering, braking, speed, or other operational parameters of the vehicle **12** to slow or stop movement of the vehicle **12**. In the example illustrated in FIG. **10**, the control circuitry **44** may determine that the driveway is too steep and communicate an instruction to adjust the brakes of the vehicle **12** to limit movement of the vehicle **12** up the driveway. Similarly, and with reference to FIG. **11**, variations in elevation, such as a sharp decline, may be detected by the control circuitry **44** based on the positional information. Further, the fields of view **34**, **36**, **42** of the RADAR modules **24**, **26**, **28** may reveal the sharp change in elevation behind the vehicle **12**. The control circuitry **44** may therefore communicate instructions to engage the brakes at the brake system **92** or adjust steering or speed of the vehicle **12**, via communication with the powertrain. In this way, the detection system **10** may limit contact with obstructions **30** (e.g., a sloped driveway) or poignant or unexpected terrain shifts. In the previous examples demonstrated in FIGS. **8** and **9**, parking maneuvering may be enhanced by the detection system **10** by indicating to the user or automatically adjusting movement of the vehicle **12** to account for the available spaces within the boundary box **196**.

(74) With continued reference to FIGS. **10** and **11**, the vehicle **12** may be the truck **126** that may selectively incorporate a cover **128** to cover the bed **124** of the truck **126**. As previously described, a sensor may detect the presence or absence of the cover **128** to allow the control circuitry **44** to define or limit the available space above the bed **124** of the truck **126**. Accordingly, the present detection system **10** may account for modifications and vehicle structure to dynamically enhance the voids **106** about the vehicle **12** and perform accurate estimations for space availability.

(75) Referring now to FIGS. **12-16C**, the detection system **10** may provide for enhanced monitoring and stability detection for the towable device **130** (e.g., trailer, another vehicle, recreational vehicle, dune buggy, camper) by employing the present time-of-flight devices **14** to monitor structures of the towable device **130** and the vehicle-to-device connection. For example, the third RADAR module **28** may be configured to gather positional data about a hitch ball of the vehicle **12** and a hitch of the towable device **130**. Other features of the vehicle-to-device connection may include a sway bar, the frame **152** of the towable device **130**, a support leg **206** for the towable device **130**, or other structural features at the vehicle-to-device connection. Based on this positional information, the control circuitry **44** may communicate instructions to the powertrain **54** and/or the brake system **92** to adjust movement of the vehicle **12**. In addition, or in an alternative, the control circuitry **44** may communicate instructions to the HMI **177** to present indications of a yaw angle of the towable device **130** relative to the vehicle **12**, a position of the towable device **130** relative to the vehicle **12**, or other information related to the stability of the towable device **130**, as will be described further herein. It is contemplated that, although illustrated as a trailer, the towable device **130** may be any apparatus configured to couple, or hitch, to the vehicle **12** to allow the vehicle **12** to tow the towable device **130**.

(76) In some examples, the positional information gathered by the RADAR modules **24**, **26**, **28** may include three-dimensional positional information of the frame **152** of the towable device **130** relative to the hitch ball and/or the tailgate **56** of the vehicle **12**. Accordingly, the yaw angle may be determined based on triangulation of the data captured from at least a pair of the plurality of RADAR modules **24**, **26**, **28**. Further, the positional information gathered by the RADAR modules **24**, **26**, **28** may be incorporated with the control circuitry **44** in combination with video data captured by a camera directed toward the vehicle-to-device connection to provide enhanced machine learning and/or artificial intelligence features for detecting the yaw angle and/or a rate of change of the yaw angle. Accordingly, by synthesizing video feedback with positional information from RADAR, various guides, such as stickers, or other markings applied to the vehicle-to-device connection may be omitted.

(77) In general, the detection system **10** may be employed to estimate a width of the towable device **130**, the length of the towable device **130**, the pitch of the towable device **130**, and any other geometric feature within the fields of view **34**, **36**, **42** of the RADAR modules **24**, **26**, **28**. The

detection system **10** may further provide for redundancy by employing the first and third RADAR modules **24**, **28** to map a left side of the towable device **130** and the second and third RADAR modules **26**, **28** to map a right side of the trailer **130**. Accordingly, side edges **208**, **210** of the front wall **138** of the towable device **130** may be detected by the RADAR modules **24**, **26**, **28** and the angle of the side edges **208**, **210** may be determined by the control circuitry **44** to determine the pitch of the towable device **130**. For example, three-dimensional positional information of the front wall **138** of the towable device **130** at various heights behind the vehicle **12** may be captured by operating the third RADAR module **28** along both the first and second angular ranges **182**, **184** previously described with respect to FIG. 6C. By gathering positional information of the front wall **138** of the trailer **130** at various heights, the pitch of the trailer **130** may be defined by the control circuitry **44**. For example, if the side edges **208**, **210** taper away from a vertical plane toward a top end **212** of the trailer **130** and are nearer to the vertical plane or nearer to the RADAR modules **24**, **26**, **28** at a bottom end **214** of the trailer **130**, the control circuitry **44** may determine that the trailer **130** is angled downwardly relative to the vehicle **12**. For example, the trailer **130** may be lifted from a front end of the trailer **130** to dump contents of the trailer **130** out of a back end of the trailer **130** or the trailer **130** may be on a decline relative to the driving surface **144** of the vehicle **12**.

(78) Referring now more particularly to FIGS. **14** and **15**, the RADAR modules **24**, **26**, **28** may be configured to map the contour of the floor **142** and/or the front wall **138** of the trailer **130** to determine the tilt **131** or pitch of the trailer **130**. For example, when the trailer **130** is a dump trailer **150** and the lift unit **153** drives the trailer **130** to an unloading position, the floor **142** of the trailer **130** may extend at an oblique angle relative to the driving surface **144**. Accordingly, the trailer **130** may be pitched upwardly at an oblique angle. As demonstrated in FIG. **14**, the tilt **131** of the floor **142**, which may be the contour, may in some examples, extend at an angular range of between zero and 180°. The threshold angle **132** may be between zero and 180°. In some examples, the threshold angle **132** is between 30° and 90°. In other examples, the threshold angle **132** is less than or more than 90°. In general, the threshold angle **132** may be determined by the system based on the length of the trailer **130**, the height of the trailer **130** (e.g., as determined based on the positional information), or other geometric parameters of the trailer **130** which may be manually entered at the user interface or may be detected by the control circuitry **44**. For example, if the height and the length of the trailer **130** are determined to be approximately equal, the threshold angle **132** may be less than if the length of the trailer **130** is greater than the height of the trailer **130**. Further, as previously described, the weight of the trailer **130**, including the contents of the trailer **130**, may be factored into determination of the controller **162** of the threshold angle **132**.

(79) Still referring to FIGS. **14** and **15**, when the trailer **130** exceeds the threshold angle **132**, the controller **162** may determine an unstable condition or a max travel of the trailer **130** to limit tipping backward of the towable device **130**, a pulling force of the vehicle **12** backward, or another unstable condition. In response, the control circuitry **44** may communicate an instruction to interrupt the lift unit driving the towable device **130** upwardly.

(80) Referring now to FIGS. **16A-16C**, various examples of unstable conditions are presented. With reference to FIG. **16A** particularly, the unstable condition corresponds to the trailer **130** exceeding the threshold angle **132** calculated by the control circuitry **44** or manually entered by the user at the user interface. In this example, an unloading operation of the trailer **130** may be interrupted by the controller **162** stopping, or communicating an instruction to stop, actuation of a motor **176** or other part of the lift unit **153** driving lifting of the trailer **130**. In this example, the RADAR modules **24**, **26**, **28** may gather positional information of the floor **142** of the trailer **130** to determine the tilt **131** of the trailer **130**.

(81) Referring particularly to FIGS. **16B** and **16C**, the tilt **131** of the trailer **130** may be determined based on the contour of the front wall **138** of the trailer **130**. In these examples, the driving surface **144** is declined (FIG. **16B**) or inclined (FIG. **16C**) resulting in unstable conditions for the trailer **130** relative to the vehicle **12**. For example, in these examples, the control circuitry **44** may

estimate the floor **142** of the trailer **130** to be generally orthogonal relative to the front wall **138**, and therefore calculate the pitch of the trailer **130** based on the projected orientation/position of the floor **142**. Thus, due to the distance of measurement provided by the RADAR modules **24**, **26**, **28** of the present disclosure, the tilt **131** or pitch of the trailer **130** may be detected in various unstable conditions.

(82) As previously described, it is contemplated that the control circuitry **44** may fix or calculate the threshold angle **132** for the trailer **130** based on the dimensions of the trailer **130** (e.g., the height, length, width, etc.). Further, weight information from the weight sensor **154** may be factored into calculation of the threshold angle **132**. Accordingly, the present detection system **10** may be provided for enhanced stability control and maneuvering of the vehicle **12**. Further, by providing communication with the mobile device **170** of the user at the rear **18** of the vehicle **12**, the user may be located relative to the three RADAR modules **24**, **26**, **28** and allow for limited obstructions **30** along the movement direction.

(83) The present detection system **10** may have other enhancements relative to imagers or camera systems for monitoring the region exterior **32** to the vehicle **12** and behind the vehicle **12**. For example, environmental challenges, such as mud, snow, ice, fogging of lenses, or any other weather-related challenge may be limited from influencing the positional information captured by the RADAR modules **24**, **26**, **28**. Accordingly, the detection system **10** may allow for reduced or omitted ultrasonic sensing typically incorporated for autonomous control and detection of objects **50**, **78**, **80** or obstructions **30** around the vehicle **12** and may limit the need or enhance the use of a camera focused toward the region exterior **32** and behind the vehicle **12**.

(84) It is to be understood that variations and modifications can be made on the aforementioned structure without departing from the concepts of the present disclosure, and further it is to be understood that such concepts are intended to be covered by the following claims unless these claims by their language expressly state otherwise.

Claims

1. A vehicle, comprising: a plurality of sides each having a left portion and a right portion, the plurality of sides including a front side, a rear side, a left side, and a right side; a plurality of RADAR modules arranged at a common side of the plurality of sides and configured to detect an obstruction in a region exterior to the vehicle, the plurality of RADAR modules including: a first RADAR module on the left portion of the common side and having a first field of view; a second RADAR module on the right portion of the common side and having a second field of view; and a third RADAR module between the first and second RADAR modules and spaced from the first RADAR module by a first distance and spaced from the second RADAR module by a second distance, the third RADAR module having a third field of view at least partially overlapping with each of the first and second fields of view; and control circuitry in communication with the plurality of RADAR modules and configured to triangulate a position of the obstruction using a first pair of the plurality of RADAR modules, wherein the third RADAR module is vertically spaced from the first and second RADAR modules.
2. The vehicle of claim 1, wherein the first distance is between a quarter of and four times the second distance.
3. The vehicle of claim 1, wherein the first distance is between a half of and twice the second distance.
4. The vehicle of claim 1, wherein the first RADAR module is vertically aligned with the second RADAR module.
5. The vehicle of claim 1, further comprising: a first taillight assembly disposed at the left portion of the rear side; and a second taillight assembly disposed at the right portion of the rear side, wherein the first and second RADAR modules are disposed in the first and second taillight

assemblies, respectively.

6. The vehicle of claim 5, wherein the control circuitry is configured to selectively execute: a blind-spot detection algorithm to determine at least one object in a blind spot for the vehicle based on the positional data captured in the first and second fields of view; and a cross-traffic detection algorithm to determine the at least one object in a cross-traffic area based on the positional data captured in the first and second fields of view.

7. The vehicle of claim 6, further comprising: a powertrain of the vehicle in communication with the control circuitry, wherein the selective execution of the blind-spot detection algorithm and the cross-traffic detection algorithm is based on a gear status of the powertrain.

8. The vehicle of claim 7, wherein the control circuitry is configured to: execute at least one of the blind-spot detection algorithm and the cross-traffic detection algorithm when the gear status is not park.

9. The vehicle of claim 7, wherein the control circuitry is further configured to execute a parking assist algorithm that guides steering of the vehicle based on positional data captured in both the third field of view and at least one of the first and second fields of view.

10. The vehicle of claim 9, wherein the control circuitry is configured to select between the parking assist algorithm and the blind-spot detection algorithm based on the gear status.

11. The vehicle of claim 10, further comprising: a tailgate at the rear side of the vehicle, wherein the third RADAR module is operably coupled with the tailgate.

12. The vehicle of claim 11, wherein the control circuitry is configured to execute a gesture recognition algorithm that determines a gesture of a user outside of the vehicle based on the positional information captured in the third field of view.

13. The vehicle of claim 12, wherein the control circuitry is configured to execute the gesture recognition algorithm when the gear status is park.

14. The vehicle of claim 1, wherein the control circuitry is configured to: triangulate the position of the obstruction using a second pair of the plurality of the RADAR modules; compare the triangulation based on the first pair of the plurality of RADAR modules to the triangulation based on the second pair of the plurality of RADAR modules; and update the position based on the comparison.

15. A vehicle, comprising: a front, a rear, a driver side, and a passenger side; a plurality of RADAR modules arranged at the rear of the vehicle and configured to detect an obstruction in a region exterior to the vehicle, the plurality of RADAR modules including: a first RADAR on the driver side of the vehicle and having a first field of view; a second RADAR module on the passenger side of the vehicle and having a second field of view; and a third RADAR module between the first and second RADAR modules and spaced from the first RADAR module by a first distance and spaced from the second RADAR module by a second distance, the third RADAR module having a third field of view at least partially overlapping with each of the first and second fields of view behind the vehicle; and control circuitry in communication with the plurality of RADAR modules and configured to triangulate a position of the obstruction using a first pair of the plurality of RADAR modules, wherein the first distance is between a half of and twice the second distance, wherein the third RADAR module is vertically spaced from the first and second RADAR modules.

16. The vehicle of claim 15, wherein the control circuitry is configured to: triangulate the position of the obstruction using a second pair of the plurality of the RADAR modules.

17. The vehicle of claim 16, further comprising: a first taillight assembly disposed at the rear on the driver side; and a second taillight assembly disposed at the rear on the passenger side, wherein the first and second RADAR modules are disposed in the first and second taillight assemblies, respectively.

18. The vehicle of claim 17, wherein the control circuitry is configured to selectively execute: a blind-spot detection algorithm to determine at least one object in a blind spot of at least one of the driver side and the passenger side based on positional data captured in the first and second fields of

view; and a cross-traffic detection algorithm to determine the at least one object in a cross-traffic area based on the positional data captured in the first and second fields of view.

19. The vehicle of claim 18, further comprising: a powertrain of the vehicle in communication with the control circuitry, wherein the selective execution of the blind-spot detection algorithm and the cross-traffic detection algorithm is based on a gear status of the powertrain.

20. A vehicle, comprising: a front, a rear, a driver side, and a passenger side; a first taillight assembly disposed at the rear on the driver side; and a second taillight assembly disposed at the rear on the passenger side; a plurality of RADAR modules arranged at the rear of the vehicle and configured to detect an obstruction in a region exterior to the vehicle, the plurality of RADAR modules including: a first RADAR module in the first taillight assembly and having a first field of view; a second RADAR module in the second taillight assembly and having a second field of view; and a third RADAR module disposed centrally between the first and second RADAR modules, the third RADAR module having a third field of view at least partially overlapping with each of the first and second fields of view behind the vehicle; and control circuitry in communication with the plurality of RADAR modules and configured to triangulate a position of the obstruction using a pair of the plurality of RADAR modules, wherein the third RADAR module is vertically spaced from the first and second RADAR modules.
